

Hedge Overlay Recommendation

model_v4 — "up-shock + vol-surge" 1-day semis hedge

Standalone overlay (src/hedge_overlay.py); the model is NOT modified.

THE RULE (decide at close T, hold T+1 only, exit next close)

FIRE if: QQQ 1-day return $\geq +2\%$ AND $z(\text{SPY 5d realized vol, 63d}) \geq 1.0$
THEN buy: short-SMH ~30% notional (default) OR SOXS ~10% (matched -0.30 beta)

WHY IT WORKS

- Targets the Jun-16 pattern: a sharp up day on surging vol -> next-day pullback.
- No single factor works alone (a QQQ up-day alone BOUNCES); the edge is the INTERACTION of up-shock x elevated vol.
- Best supporting signal (sensitivity): 5-day realized-vol z-score. Relative vol beats absolute level / surge-ratio; the 5d window beats 1d and 20d.

QUANTIFIED VALUE (\$100k book, SOXS @10%, 68 fires)

1-DAY hold (THE RULE): net \$+12,348 [wins \$+30,812 / losses \$-18,464, 54% win]
if held 5D: \$-178 if held 20D: \$-86,752 -> the 1-day exit IS the edge
best saves: 2020-03-13 \$+4,526 (book -9.0%) - 2026-06-16 \$+1,622 (book -3.8%)

RISK-ADJUSTED (default short-SMH 0.30)

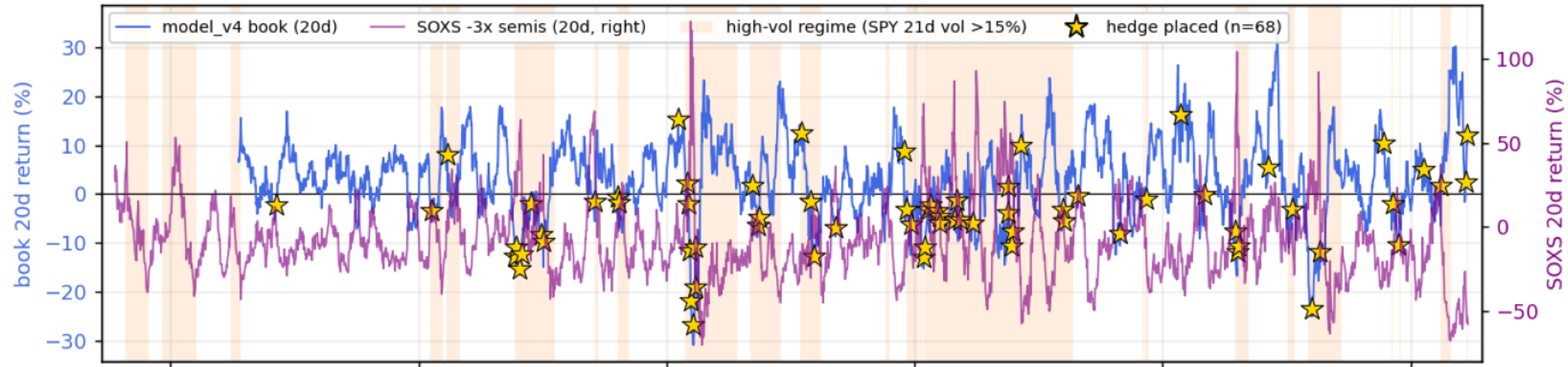
FULL: dSharpe +0.06 sleeve PnL +11.9% worst on-day -9.0% -> -4.7%
OOS (≥ 2024): dSharpe +0.03 sleeve +1.7%
Walk-forward: 6/10 active years dSharpe >0 ; gains concentrated in vol-spike years

CAVEATS

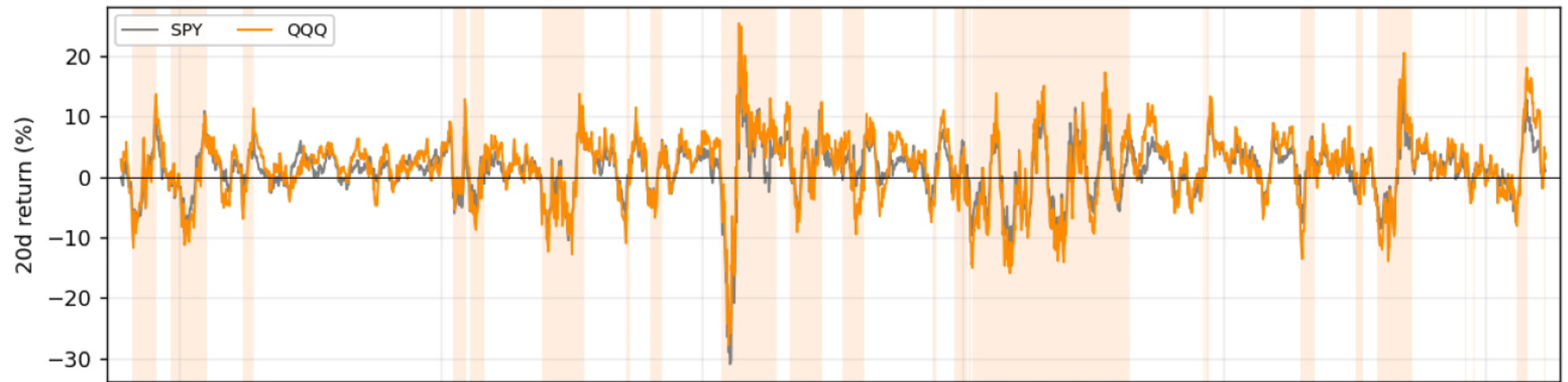
- A Sharpe-adder + single-day-blow softener, NOT an all-time-max-drawdown reducer.
- Edge concentrates in vol spikes; marginal in calm years; slightly -ve in 2022 bear.
- SOXS spread/decay understated by flat 10bp; short-SMH is cheaper & decay-free.
- Paper / recommendation only. Re-validate after any model or universe change.

SOXS hedge @10% on \$100k book — PnL by hold horizon (n=68 fires)
1-DAY (the rule): net \$+12,348 [wins \$+30,812 / losses \$-18,464, 54% win]
5-DAY: \$-178 20-DAY: \$-86,752 -> strict 1-day exit is what saves \$
best 1D saves: 2020-03-13 \$+4,526 (book -9.0%) - 2026-06-16 \$+1,622 (book -3.8%)

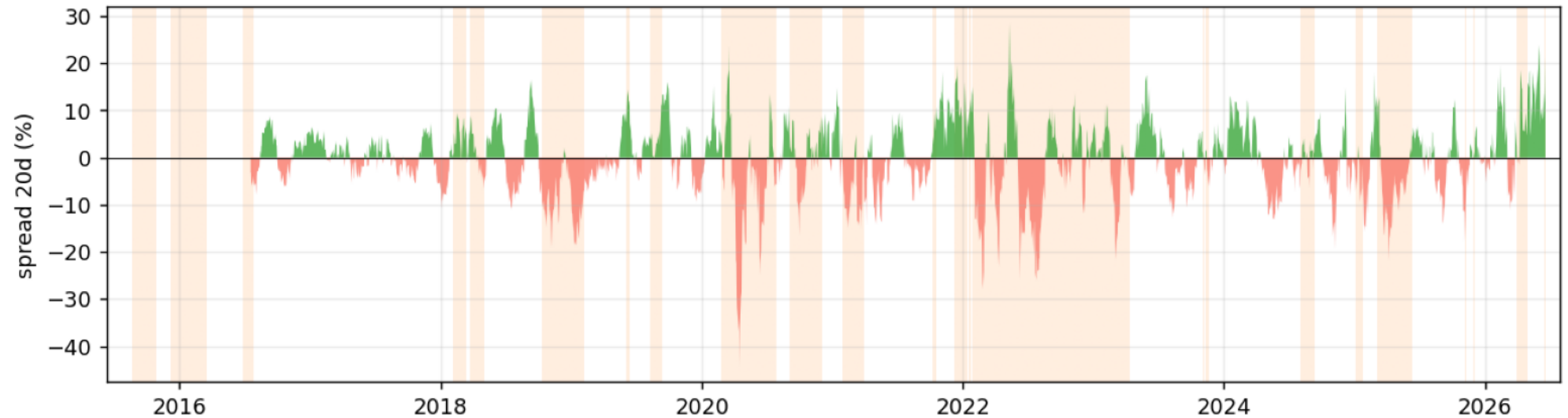
model_v4 portfolio vs SOXS hedge — ★ = hedge placed (up-shock + vol-surge)



SPY / QQQ — trailing 20-day return



Top10–Bottom10 spread (20d, fill)



Backtest — windows & yearly walk-forward

Default config: short-SMH 0.30, trigger qqq_1d>=2%% & z(spy_vol_5d)>=1. dSh = hedged minus book Sharpe; dDD = max-drawdown change (+ = better).

WINDOW SUMMARY

window	days	on		bookSh	hedSh	dSh		bookDD	dDD		onBook%	paid%	sleeve%
FULL	2513	68		+1.24	+1.30	+0.06		-39.5%	-0.4%		-0.44	56	+11.91%
COVID	1561	47		+1.22	+1.23	+0.01		-39.5%	-0.4%		0.07	49	+0.19%
TRAIN(<oos)	1896	52		+1.18	+1.24	+0.07		-38.1%	+0.8%		-0.58	60	+10.17%
OOS	617	16		+1.42	+1.46	+0.03		-39.5%	-0.4%		-0.00	44	+1.73%

YEARLY WALK-FORWARD

year	days	on		bookSh	hedSh	dSh		bookDD	dDD		onBook%	paid%	sleeve%
2016	136	1		+2.73	+2.72	-0.02		-6.0%	-0.0%		-0.17	0	-0.15%
2017	251	0		+1.82	+1.82	+0.00		-10.8%	+0.0%		na	na	+0.00%
2018	251	8		+0.71	+0.73	+0.02		-28.9%	-0.7%		-0.69	62	+0.29%
2019	252	4		+2.53	+2.60	+0.06		-10.1%	+1.2%		-0.80	75	+0.73%
2020	253	11		+1.44	+1.77	+0.33		-33.8%	+6.7%		-2.03	73	+10.01%
2021	252	7		+0.49	+0.53	+0.04		-21.6%	+1.2%		-0.41	71	+1.16%
2022	251	16		-0.76	-0.82	-0.06		-31.8%	+0.1%		0.15	50	-1.17%
2023	250	5		+2.23	+2.21	-0.02		-16.8%	-0.2%		0.33	40	-0.69%
2024	252	6		+2.12	+2.14	+0.02		-20.4%	+1.5%		0.60	50	+0.10%
2025	250	6		-0.15	-0.08	+0.07		-37.1%	-0.3%		-0.44	50	+2.16%
2026	115	4		+3.01	+3.01	-0.00		-12.1%	+0.0%		-0.24	25	-0.52%

Source: backtests/hedge_overlay_model_v4_summary.md | panel model_v4_timeseries.csv (2016-06-20 to 2026-06-18)